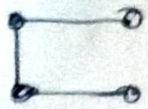
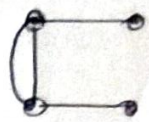
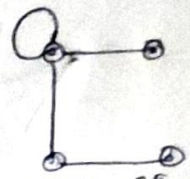
1. <u>Simple Graph</u>: A graph in which each edge connects two different vertices and, no two edges connects same pair of vertices.



Simple graph

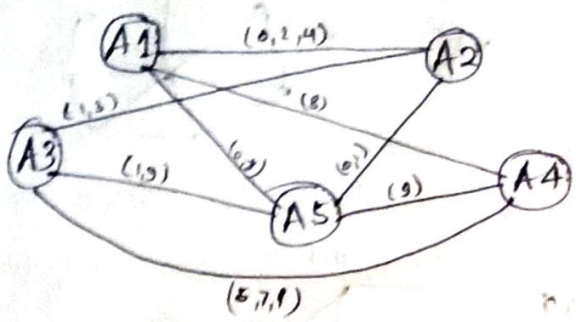


Not simple



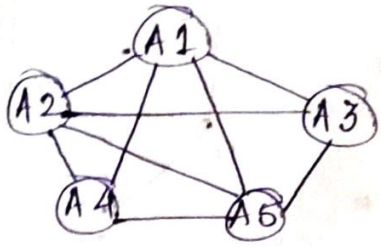
Not Simple [loop]

2. (a)

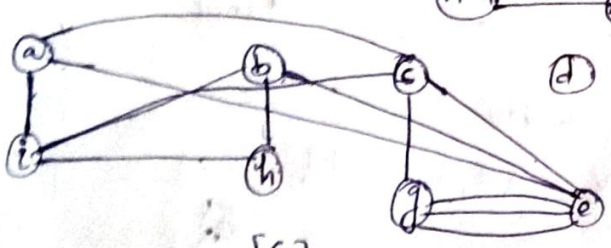


intersection graph of set (A1, A2, A3, A4, A5)

3. (b)



3.



[G]

~~deg(G)~~

number of vertices = 8

number of edges = 12

deg(a) = 3 deg(b) = 2 deg(c) = 4 deg(d) = 0

deg(i) = 3 deg(h) = 2 deg(g) = 4 deg(e) = 6

isolated vertice = {d}

pendent vertice = { }

4. Consider a simple graph of vertex n [≥ 2] maximum degree of any vertex is [$n-1$]. A vertex can have any degree from [0, $n-1$]

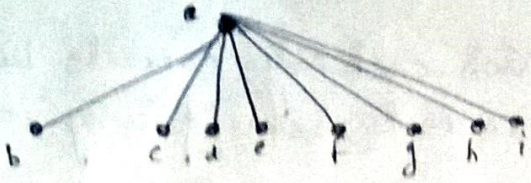
According to pigeonhole principle, there must be atleast two vertices that must have same ~~num~~ degree.

Hence, due to pigeonhole principle, for a simple graph there must be two vertices with same degree

5. K_7

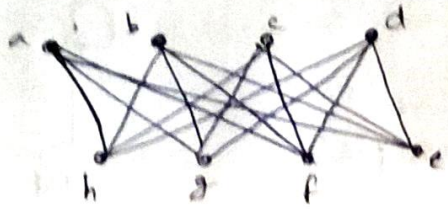
all possible edges for a simple graph of vertice '7'.

$K_{1,8}$

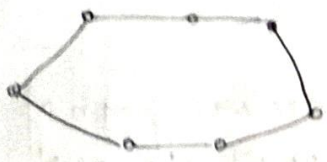


A bipartite complete graph with one (1) ~~edge~~ vertex and (8) ~~ede~~ vertex.

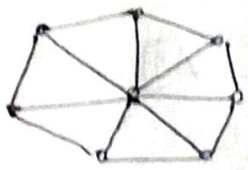
$K_{4,4}$



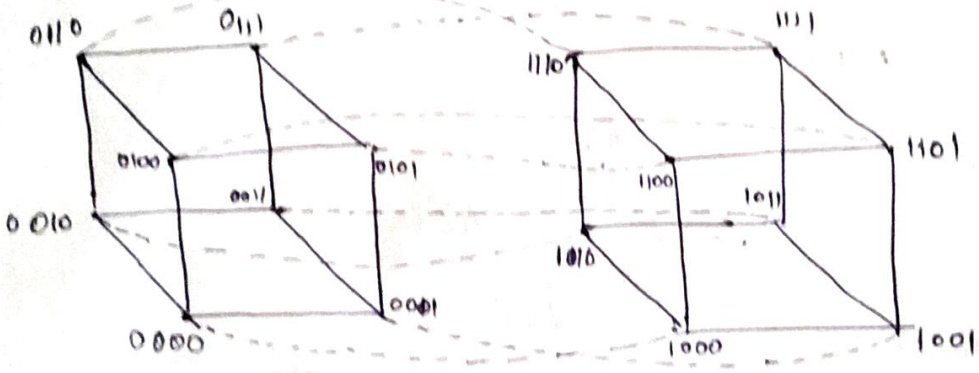
$C_7 \rightarrow$



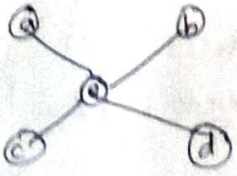
$W_7 \rightarrow$



$Q_4 \rightarrow$

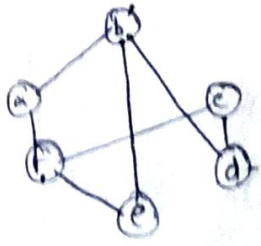


6.



← bipartite Graph with $V_1 = \{e\}$ $V_2 = \{a, b, c, d\}$

not a bipartite Graph ~~with~~ ~~$V_1 = \{b, d, f\}$ $V_2 = \{a, c, e\}$~~ →



7. <u>Hall's Marriage Problem</u>: The bipartite graph $G = (V, E)$ with bipartition (V_1, V_2) has a complete matching from V_1 to V_2 if and only if $|N(A)| \geq |A|$ for all subsets A of V_1. It states that, a matching between two bipartite subsets is possible if and only if, for every subset ~~$X \subseteq B$~~ $X \subseteq V_2$

8. Let V_1 = Set of men on island ; $N(V_1) = A$
V_2 = set of women on island ; $N(V_2) = B$

According to question,

clearly, for V_1

$$Ak \geq A$$

i.e. A perfect maching exist for the given condition, due to Hall's Theorem.

Same goes for V_2

$$Ak \geq A$$

~~it.~~ ~~t~~

Clearly from above, if a perfect match is possible from both sides [i.e. V_1 & V_2], then there is way through which everyone will get married to someone they wish to marry.

9. (a) K_n : A complete graph with 'n' vertices
$^nC_2 \left(= \frac{n(n-1)}{2}\right)$ edges.

(b) C_n : A cycle has 'n' vertices and 'n' edges.

(c) W_n : A wheel has $(n+1)$ vertices and $n+n\ (=2n)$ edges

(d) $K_{m,n}$: A complete bipartite graph with $N(V_1)=m$ $N(V_2)=n$
vertices $= (m+n)$
edges $= m * n$

(e) Q_n : n-hypercube has 2^n vertices.
edges $=$ ~~2^n~~
$E(n) = 2E(n-1) + 2^{n-1}$ (recursive function)

10. K_3 has 3 vertices and 3 edges.

Case I : All three vertices are chosen
then number of subgraphs $=$ ~~$^3C_3 = 3!$~~ $2\times2\times2 = 8$

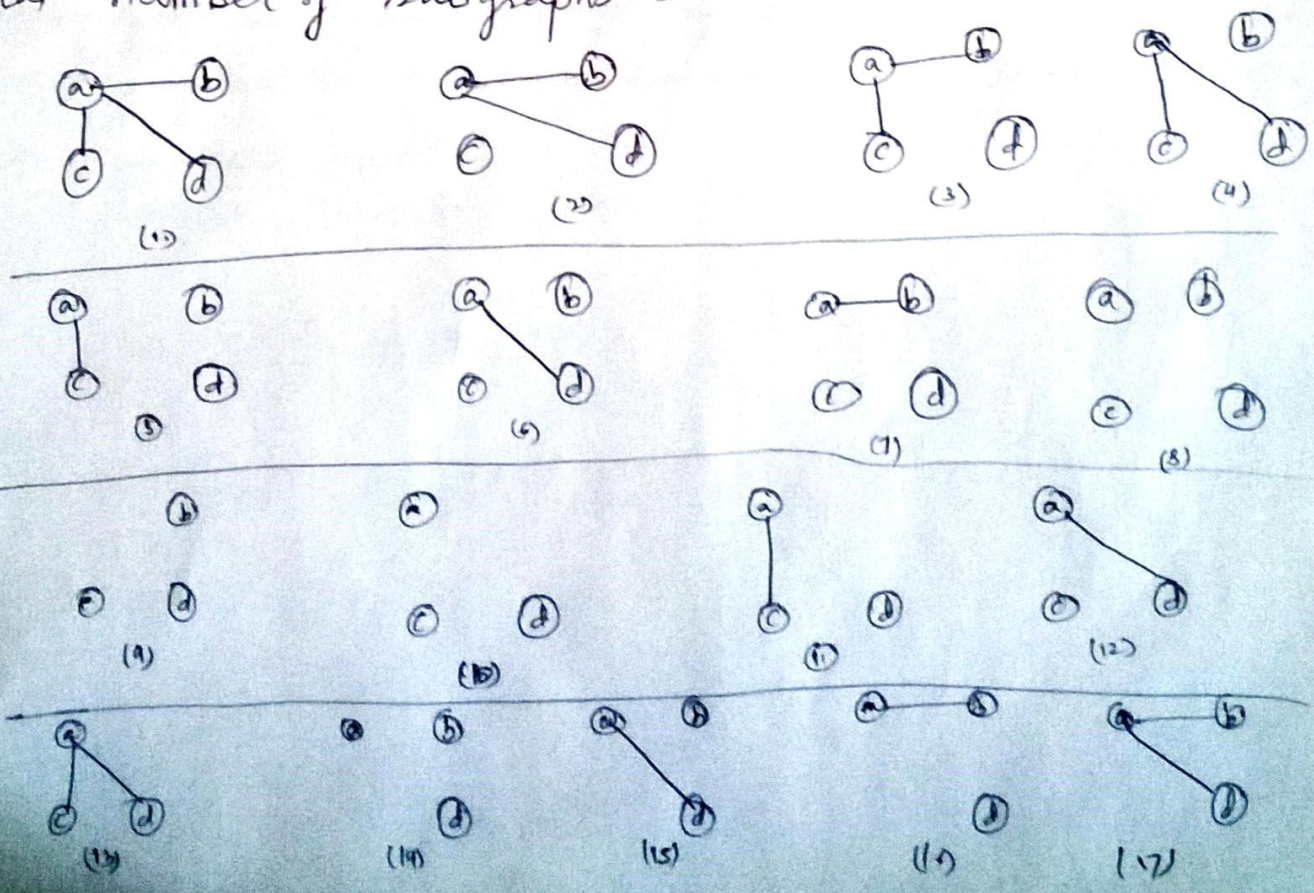
Case II : Any two vertices are chosen
then number of subgraphs $= {}^3C_2 \times 2 = 6$

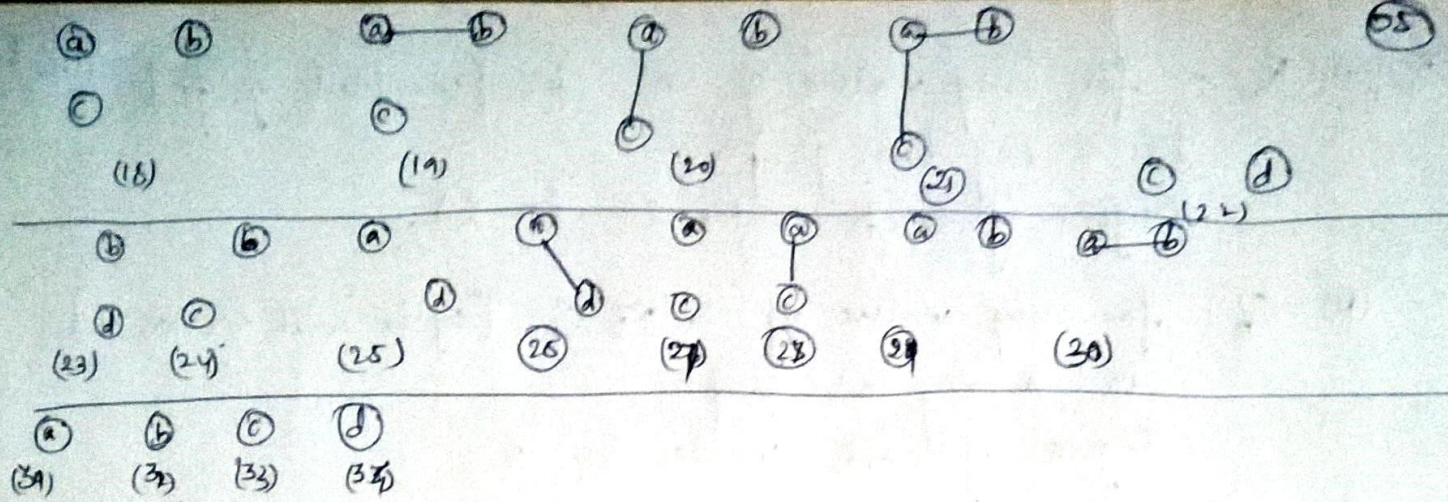
Case III : Any one vertex is chosen
then number of subgraphs $= 3$

~~Case IV : No vertex is chosen~~
~~No of subgraphs = 1~~

Total number of subgraphs $= 8+6+3 = 17$

11.





A total of 33 (and a null graph) can be made out of given graph. i.e. 33 subgraphs.

12. Number of vertices V
Number of edges e
$m \leftarrow$ minimum degree $\quad M \leftarrow$ maximum degree

<u>To show</u>

$$\frac{2e}{V} \leq M \qquad \frac{2e}{V} \geq m$$

In a graph, we have $\quad 2e = \sum_{v \in V} \deg(v)$

So,

$$\frac{2e}{V} = \frac{\sum_{v \in V} \deg(v)}{V}$$

also, we know that ~~average of~~ arithmatic mean of 'n' numbers is greater than equal to smallest number and smaller than equal to greatest number,

here, smallest number $= m$
largest number $= M$

So,

$$m \leq \frac{2e}{V} \leq M$$

or

$$m \leq \frac{2e}{V} \quad \text{and} \quad \frac{2e}{V} \leq M$$

13[a](a) K_n : for any value of 'n' K_n [complete graph] is a regular graph.
Degree of each vertex is $(n-1)$.

(b) C_n : for any value of 'n' ≥ 3 C_n [cycle with vertex n] is a regular graph.
Degree of each vertex is 2.

(c) W_n : ~~for n=3~~ A wheel is made by introducing a vertex, thus increasing degree of each existing vertex to 3.
$\therefore$ for $n = 3$, i.e. W_3 will ~~be~~ a regular ~~vertex~~ graph.

(d) Q_n : n-hypercube is regular for any value of 'n'.
Degree of each vertex depends on n as 'n'.

[b] $K_{m,n}$ a complete bipartate graph can only be regular when any one of the following is true

(i) $m = 0$; or
(ii) $n = 0$; or
(iii) $m = n$

[c] if all vertices have degree 4, then total number of edges can be calculated as

$$\frac{4n}{2} = e$$

Also, $e = 10$, then

$$\frac{4 \times n}{2} = 10 \Rightarrow n = 5$$

$\therefore$ The regular graph has five (5) vertices. [K_5]